

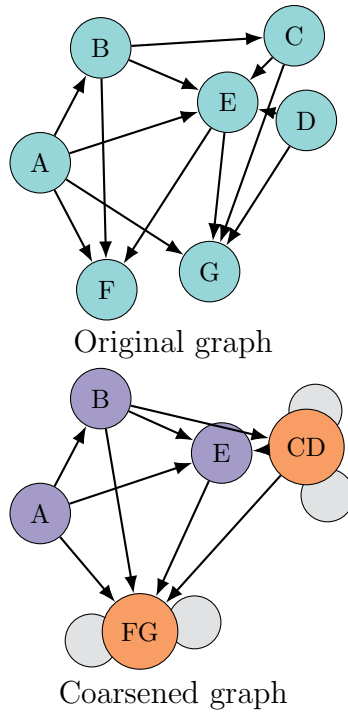


Figure 1: Illustration of the graph coarsening procedure. Nodes  $C$  and  $D$  are merged into super-node  $CD$ , and nodes  $F$  and  $G$  into super-node  $FG$ .

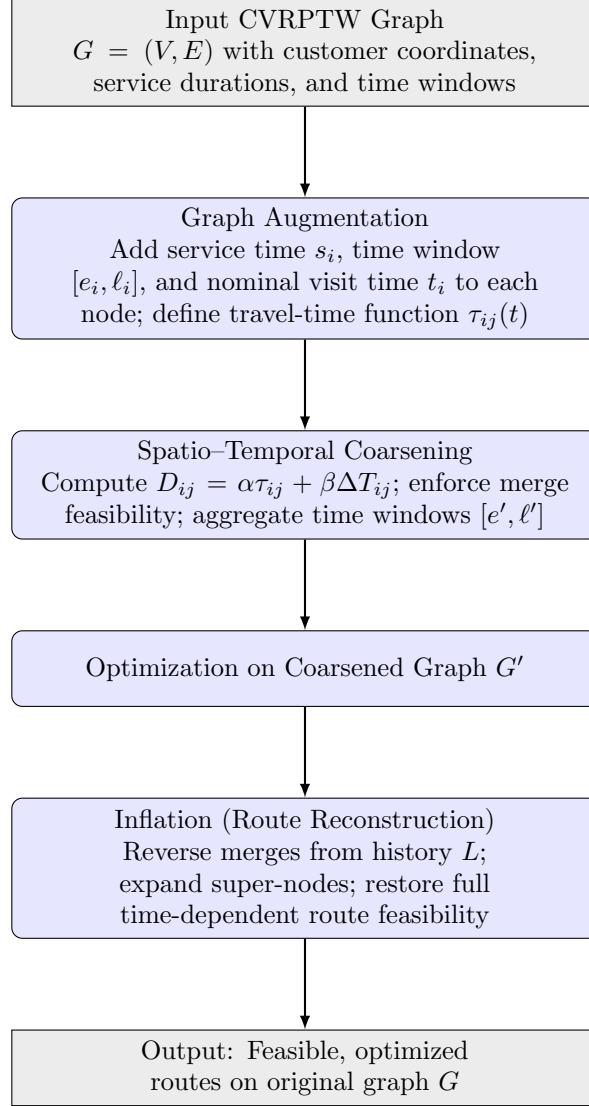
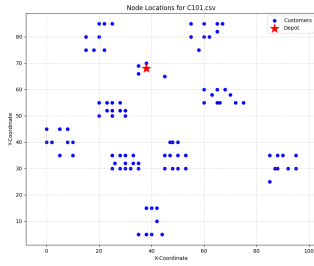
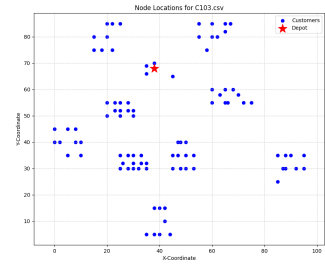


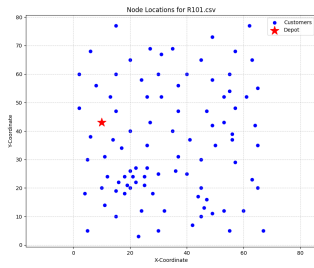
Figure 2: Overall workflow of the proposed spatio-temporal graph coarsening framework for CVRPTW. The pipeline proceeds top-down: from graph augmentation and coarsening, through reduced-graph optimization, to route inflation and feasibility restoration.



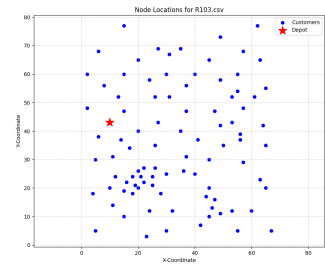
(a) C101



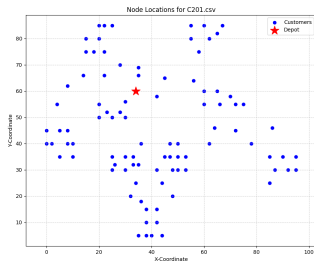
(b) C103



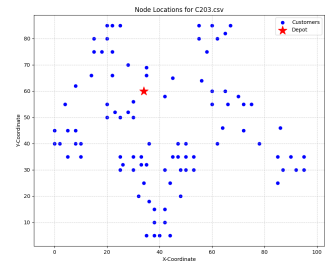
(c) R101



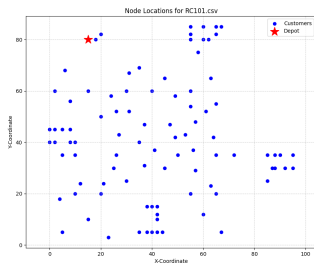
(d) R103



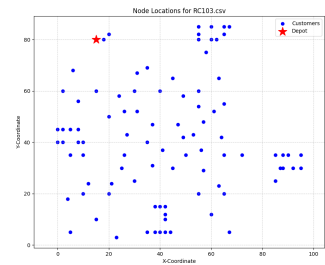
(e) C201



(f) C203



(g) RC101



(h) RC103

Figure 3: A subset of instances of the Solomon dataset used for the experiments. For clarity, edges are omitted.